

Billy Lukito Danuharja

Electrical Engineer

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SUMMARY

Computer Engineering undergraduate with a strong interest in robotics with hands-on experience building systems that span across air and land. Throughout my academic and project journey, Experience working on a variety of robotic platforms, including quadcopters, fixed-wing drones, and ground-based mobile robots. These projects have allowed me to explore and integrate complex systems involving control, sensing, and autonomy across different environments. I have a strong foundation in embedded systems and low-level programming, with a passion for turning hardware into intelligent, responsive machines. Whether it's navigating flying autonomous drones or designing smart autonomous car, I'm driven by curiosity and a deep interest in building reliable, real-world robotic solutions.

EDUCATION

Bachelor of Applied Engineering, Computer Engineering

Politeknik Elektronika Negeri Surabaya

GPA: 3.71 / 4.00

2022 – Present
Surabaya, Indonesia

INTERNSHIP EXPERIENCE

PT. Piranti Kecerdasan Buatan

Research and Development

Aug 2025 – Dec 2025
Tangerang, Indonesia

- Developed an ACC system for a 1:10 scale Ackerman mobile robot to maintain safe distance and stable speed against a preceding obstacle/vehicle.
- Integrated LiDAR distance measurement and encoder feedback for real-time speed and distance monitoring.
- Designed and evaluated FSFM-based control logic with PSO optimization for improved safety-distance response.

PT. Infoglobal Teknologi Semesta

Electronic Development Engineer

Jan 2025 – Jun 2025
Surabaya, Indonesia

- Designed and developed a DVR Reader board from the ground up, including schematic capture and PCB layout using KiCad.
- Developed embedded firmware, supported migration from ATmega328 to STM32, and performed board bring-up, hardware-software integration, functional testing, debugging, and technical documentation until the system was operational.
- Performed hardware debugging, system troubleshooting, validation checks, and technical documentation/reporting.

TEAM AND ORGANIZATION EXPERIENCE

Pekan Kreativitas Mahasiswa - Karsa Cipta 2025

Leader and Electrical Division

Jul 2025 – Nov 2025
Surabaya, Indonesia

- Led the development of an autonomous firefighting drone prototype as Team Leader and Electrical Division member.
- Designed and integrated electrical wiring for the flight controller, extinguisher actuator, and computer vision hardware.
- Performed system troubleshooting and coordinated hardware-software integration for safe flight testing.

Cakrawala Skala - PENS UAV Student Research Team

Co-Leader and Head of Electrical Division

Oct 2024 – Oct 2025
Surabaya, Indonesia

- Led the electrical division in developing UAV/autonomous robot systems, managing technical planning, task coordination, and integration readiness.
- Supervised electrical wiring, hardware assembly, functional testing, and troubleshooting during system development.
- Helped lead the team to 2nd place at SAFMC 2025 D1 Man-Machine in Singapore, outperforming teams from several Southeast Asian countries.

Lembaga Minat Bakat PENS

Student Resource Development Staff

Feb 2023 – Feb 2025
Surabaya, Indonesia

- Supported student development programs through event planning, documentation, and organizational coordination.
- Led Upgrading 2025 as Chief Organizer, coordinating committee workflow, event operations, and execution for 80 participants.
- Led Sekolah UKM 2024 and 2025 as Chief Organizer, managing program preparation, participant coordination, and execution for 24 participants.

Pekan Kreativitas Mahasiswa - Karsa Cipta 2024

Electrical Division

Jul 2024 – Nov 2024
Surabaya, Indonesia

- Designed and validated a custom Power Distribution Board (PDB) to ensure stable power delivery for the flight controller and propulsion system.
- Integrated the seed bombing actuator with the electrical system to support synchronized payload deployment during mission testing.
- Developed wiring schematics, implemented overcurrent protection, and performed troubleshooting to improve flight-test reliability.

Gakrawala Skala - PENS UAV Student Research Team

Electrical Division

Oct 2023 – Oct 2024
Surabaya, Indonesia

- Integrated microcontrollers, sensors, actuators, and motor control components for autonomous robot/UAV system development.
- Supported electrical wiring, embedded system prototyping, hardware debugging, and functional validation to improve system reliability.
- Contributed to technical preparation for SAFMC 2024 D1 Man-Machine, supporting the team's semi-finlist achievement in Singapore.

AWARDS

Project Funding Grantee Pekan Kreativitas Mahasiswa – Karsa Cipta 2025

Nov 2025

Indonesian Talent Development Center (BPTI), National Center for Excellence (Puspresnas), Ministry of Education, Culture, Research, and Technology of the Republic of Indonesia
Entitled "Design and Development of an Autonomous Quadcopter Based on Computer Vision for Fire Detection and Extinguishment in High-Rise Buildings"

2nd Place Singapore Amazing Flying Machine Competition (SAFMC) 2025

Mar 2025

DSO National Laboratories and Science Centre Singapore, in collaboration with the Ministry of Defence

Project Funding Grantee Pekan Kreativitas Mahasiswa – Karsa Cipta 2024

Nov 2024

Indonesian Talent Development Center (BPTI), National Center for Excellence (Puspresnas), Ministry of Education, Culture, Research, and Technology of the Republic of Indonesia
Entitled "Hybrid Unmanned Aerial Vehicle (UAV) for Reforestation in Extreme Terrain Using the Seed Bombing Method"

Semi-Finalist Singapore Amazing Flying Machine Competition (SAFMC) 2024

Mar 2024

DSO National Laboratories and Science Centre Singapore, in collaboration with the Ministry of Defence

SKILLS

- **Programming:** C, C++, Python
- **Embedded Systems:** Arduino IDE, STM32CubeIDE, Arduino, STM32, ATmega328
- **Robotics & Middleware:** ROS, autonomous robot system, UAV/quadcopter platform, Ackermann mobile robot, Gazebo Simulation
- **PCB & Electrical Design:** KiCad, electrical wiring, component integration, power distribution
- **Sensors & Communication:** LiDAR, encoder, UART, I2C, SPI, PWM, sensor-actuator interfacing
- **Control & Automation:** Fuzzy Logic, Social Force Model, PSO Optimization, Adaptive Cruise Control